Alpha-Long-Short Trading Strategy - Complete Beginner's Guide

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What is This Project?

Simple Explanation:

Imagine you're betting on a horse race. Instead of just betting on ONE horse to win, you: 1. **Bet FOR some horses** (you think they'll win) → This is called **LONG** 2. **Bet AGAINST other horses** (you think they'll lose) → This is called **SHORT**

This project does the same thing but with **stocks** instead of horses!

What the Project Does:

- Looks at **10 big US companies** (Apple, Microsoft, Amazon, Google, Meta, Tesla, Nvidia, JPMorgan, Johnson & Johnson, Walmart)
- Every month, it picks:
  - **Top 3 stocks** to BUY (go LONG) - stocks expected to go UP
  - **Bottom 3 stocks** to SELL (go SHORT) - stocks expected to go DOWN

Real World Example:

Let's say we have $100,000: - We spend $50,000 buying Apple, Microsoft, Nvidia (LONG) - We borrow and sell $50,000 worth of Tesla, Meta, Walmart (SHORT)

If Apple goes up 10% and Tesla goes down 10%, we make money from BOTH!

Key Terms with Formulas & Examples

1. Long Position

Meaning: Buying a stock hoping it goes UP

Formula:

```
Profit/Loss = (Selling Price - Buying Price) × Number of Shares
Return % = (Selling Price - Buying Price) / Buying Price × 100
```

Example:

```
You buy 100 shares of Apple at $150 each
Cost = 100 × $150 = $15,000

Apple rises to $170
Value = 100 × $170 = $17,000

Profit = $17,000 - $15,000 = $2,000
Return = ($170 - $150) / $150 × 100 = 13.33%
```

---

2. Short Position

Meaning: Selling borrowed stock hoping it goes DOWN

Formula:

```
Profit/Loss = (Selling Price - Buying Price) × Number of Shares
              (Note: You sell FIRST, then buy back)
Return % = (Entry Price - Exit Price) / Entry Price × 100
```

Example:

```
You borrow and sell 100 shares of Tesla at $200 each
Cash received = 100 × $200 = $20,000

Tesla drops to $160
You buy back 100 shares at $160 = $16,000

Profit = $20,000 - $16,000 = $4,000
Return = ($200 - $160) / $200 × 100 = 20%
```

If Tesla rose instead:

```
Tesla rises to $220
You must buy back at $220 = $22,000
```

```
Loss = $20,000 - $22,000 = -$2,000
```

---

3. Return Calculation

Daily Return Formula:

```
Daily Return = (Today's Price - Yesterday's Price) / Yesterday's Price

Or simply: R_t = (P_t - P_{t-1}) / P_{t-1}
```

Example:

```
Apple yesterday: $150
Apple today: $153

Daily Return = ($153 - $150) / $150 = 0.02 = 2%
```

Cumulative Return Formula:

```
Cumulative Return = (1 + R ) × (1 + R ) × ... × (1 + R ) - 1
```

Example (3 days of returns):

```
Day 1: +2%, Day 2: -1%, Day 3: +3%

Cumulative = (1.02) × (0.99) × (1.03) - 1
           = 1.0399 - 1
           = 0.0399 = 3.99%
```

---

4. Dollar-Neutral

Meaning: Equal dollar amounts in LONG and SHORT positions

Formula:

```
Long Exposure = Short Exposure
Net Exposure = Long - Short = 0
Gross Exposure = Long + Short
```

Example:

```
Portfolio: $100,000

Long positions:
  AAPL: $16,667
  MSFT: $16,667
  NVDA: $16,666
  Total Long = $50,000
```

```
Short positions:
  TSLA: -$16,667
  META: -$16,667
  WMT:  -$16,666
  Total Short = -$50,000

Net Exposure = $50,000 - $50,000 = $0   (Dollar Neutral!)
Gross Exposure = $50,000 + $50,000 = $100,000
```

---

5. Sharpe Ratio

Meaning: Risk-adjusted return (how much return per unit of risk)

Formula:

```
Sharpe Ratio = (Portfolio Return - Risk-Free Rate) / Portfolio Volatility

                    R_p - R_f
Sharpe Ratio =
                       _p

Where:
- R_p = Annualized portfolio return
- R_f = Risk-free rate (usually ~2% Treasury rate)
-  _p = Annualized portfolio volatility (standard deviation)
```

Example:

```
Portfolio annual return: 15%
Risk-free rate: 2%
Portfolio volatility: 10%

Sharpe = (15% - 2%) / 10% = 13% / 10% = 1.30

Interpretation:
- Sharpe < 0: Bad
- Sharpe 0-1: Below average
- Sharpe 1-2: Good
- Sharpe > 2: Excellent

Our 1.30 is GOOD!
```

---

6. Maximum Drawdown

Meaning: Largest drop from peak to trough

Formula:

```
Drawdown at time t = (Peak Value - Current Value) / Peak Value × 100

Max Drawdown = Maximum of all drawdowns over the period
```

Example:

```
Day 1: Portfolio = $100,000 (Peak)
Day 2: Portfolio = $95,000
Day 3: Portfolio = $85,000 (Trough)
Day 4: Portfolio = $90,000
Day 5: Portfolio = $110,000 (New Peak)
Day 6: Portfolio = $100,000

Drawdown on Day 3 = ($100,000 - $85,000) / $100,000 = 15%
Drawdown on Day 6 = ($110,000 - $100,000) / $110,000 = 9.1%

Maximum Drawdown = 15% (the biggest drop)
```

---

7. Alpha ( )

Meaning: Excess return above what's expected given market risk

Formula (CAPM-based):

```
 = R_p - [R_f +   × (R_m - R_f)]

Where:
- R_p = Portfolio return
- R_f = Risk-free rate
-   = Portfolio beta
- R_m = Market return
```

Example:

```
Portfolio return: 18%
Risk-free rate: 2%
Beta: 0.5
Market return: 12%

Expected Return = 2% + 0.5 × (12% - 2%) = 2% + 5% = 7%
Alpha = 18% - 7% = 11%

This means we generated 11% MORE than expected!
```

---

8. Beta ( )

Meaning: How much the portfolio moves with the market

Formula:

```
 = Covariance(Portfolio, Market) / Variance(Market)

     Cov(R_p, R_m)
 =
       Var(R_m)
```

Interpretation:

```
 = 1.0: Moves exactly with market
 = 0.5: Moves half as much as market
 = 0.0: No correlation with market (our goal!)
 = -0.5: Moves opposite to market
```

Example:

```
If market goes up 10%:
   = 1.0 → Portfolio goes up ~10%
   = 0.5 → Portfolio goes up ~5%
   = 0.0 → Portfolio unaffected
   = -0.5 → Portfolio goes down ~5%

Our dollar-neutral strategy targets    0
```

---

9. Z-Score Normalization

Meaning: Standardizing values to compare across different stocks

Formula:

```
Z-score = (Value - Mean) / Standard Deviation

         X -
Z =
```

Example:

```
Stock returns today:
  AAPL: +3%
  MSFT: +2%
  TSLA: -1%
  Mean = (3 + 2 - 1) / 3 = 1.33%
  Std Dev = 1.70%
```

```
Z-scores:
  AAPL: (3 - 1.33) / 1.70 = +0.98 (above average)
  MSFT: (2 - 1.33) / 1.70 = +0.39 (above average)
  TSLA: (-1 - 1.33) / 1.70 = -1.37 (below average)
```

The 3 Alpha Signals - Deep Dive

Signal 1: Momentum (Weight: 61%)

Concept: Stocks going UP tend to KEEP going up (trend following)

Formula:

```
Momentum = (Price_today - Price_20_days_ago) / Price_20_days_ago

         P_t - P_{t-20}
Mom =
           P_{t-20}
```

Then Z-score normalize across all stocks:

```
Momentum_zscore = (Mom_stock - Mean_all_stocks) / StdDev_all_stocks
```

Worked Example:

```
20 days ago → Today prices:
  AAPL: $150 → $165  | Mom = (165-150)/150 = +10%
  MSFT: $300 → $315  | Mom = (315-300)/300 = +5%
  TSLA: $200 → $180  | Mom = (180-200)/200 = -10%

Mean = (10 + 5 - 10) / 3 = 1.67%
StdDev = 8.50%

Z-scores:
  AAPL: (10 - 1.67) / 8.50 = +0.98   (Buy signal)
  MSFT: (5 - 1.67) / 8.50 = +0.39
  TSLA: (-10 - 1.67) / 8.50 = -1.37   (Short signal)
```

Signal 2: Mean Reversion (Weight: 23%)

Concept: Stocks that dropped too much will "bounce back"

Formula:

```
50-day Moving Average = Sum of last 50 prices / 50

Deviation = (Price - MA50) / MA50
```

```
Mean Reversion Signal = -Deviation  (negative because BELOW MA = BUY)
```

Worked Example:

```
Current prices and 50-day MAs:
  AAPL: Price=$165, MA50=$160 | Dev = (165-160)/160 = +3.1%
  MSFT: Price=$315, MA50=$320 | Dev = (315-320)/320 = -1.6%
  TSLA: Price=$180, MA50=$210 | Dev = (180-210)/210 = -14.3%

Mean Reversion = -Deviation:
  AAPL: -3.1% (above MA → might drop → SELL signal)
  MSFT: +1.6% (below MA → might rise → weak BUY)
  TSLA: +14.3% (way below MA → might bounce → strong BUY signal!)

After Z-score normalization, TSLA gets highest mean reversion score.
```

Signal 3: Volatility (Weight: 16%)

Concept: Stable stocks are safer and preferred

Formula:

```
Daily Returns = (P_t - P_{t-1}) / P_{t-1}

30-day Volatility = StdDev(last 30 daily returns) × √252
                    (multiply by √252 to annualize)

Volatility Signal = -Volatility  (negative because LOW vol = GOOD)
```

Worked Example:

```
30-day annualized volatility:
  AAPL: 25%
  MSFT: 22%
  TSLA: 55%
  JNJ:  15%

Volatility Signal = -Volatility:
  JNJ:  -15% → Highest score (most stable)
  MSFT: -22%
  AAPL: -25%
  TSLA: -55% → Lowest score (too jumpy)
```

Combining the 3 Signals

Formula:

```
Combined Score = (0.61 × Momentum_z) + (0.23 × MeanRev_z) + (0.16 × Volatility_z)
```

Worked Example:

```
Stock    | Momentum_z | MeanRev_z | Volatility_z | Combined Score
---------|------------|-----------|--------------|----------------
AAPL     |   +0.98    |   -0.50   |    +0.20     | 0.61(0.98) + 0.23(-0.50) + 0.16(0.20) = +(
MSFT     |   +0.39    |   +0.30   |    +0.40     | 0.61(0.39) + 0.23(0.30) + 0.16(0.40) = +0.
TSLA     |   -1.37    |   +1.20   |    -0.60     | 0.61(-1.37) + 0.23(1.20) + 0.16(-0.60) = -
JNJ      |   -0.20    |   +0.10   |    +1.00     | 0.61(-0.20) + 0.23(0.10) + 0.16(1.00) = +(

Ranking:
1. AAPL: +0.51 → LONG
2. MSFT: +0.37 → LONG
3. JNJ:  +0.06 → LONG
...
10. TSLA: -0.66 → SHORT
```

Portfolio Construction Math

Position Sizing (Dollar-Neutral Mode)

Formula:

```
Long Weight per Stock = 50% / Number of Long Stocks
Short Weight per Stock = -50% / Number of Short Stocks
```

Example with $100,000 portfolio:

```
Total Capital = $100,000

LONG side (50% = $50,000):
  3 stocks → $50,000 / 3 = $16,667 each

  AAPL: +$16,667 (buy ~101 shares at $165)
  MSFT: +$16,667 (buy ~53 shares at $315)
  JNJ:  +$16,667 (buy ~105 shares at $159)

SHORT side (50% = $50,000):
  3 stocks → $50,000 / 3 = $16,667 each

  TSLA: -$16,667 (short ~93 shares at $180)
  META: -$16,667 (short ~33 shares at $500)
  WMT:  -$16,667 (short ~100 shares at $167)
```

```
Total Long = +$50,000
Total Short = -$50,000
Net Exposure = $0
```

Position Sizing (Bull Market Mode)

When: S&P 500 Price > 200-day Moving Average

Formula:

```
Long Weight per Stock = 100% / Number of Long Stocks
Short Weight = 0% (no shorting)
```

Example:

```
SPY Price = $450, SPY MA200 = $420
Since 450 > 420 → BULL MARKET → Long-Only Mode

Total Capital = $100,000

LONG only (100% = $100,000):
  3 stocks → $100,000 / 3 = $33,333 each

  AAPL: +$33,333
  MSFT: +$33,333
  JNJ:  +$33,334

  No short positions!
```

Performance Metrics Formulas

1. Cumulative Return

```
Cumulative Return = (Final Portfolio Value / Initial Value) - 1

Example:
Initial = $100,000, Final = $165,000
Cumulative Return = ($165,000 / $100,000) - 1 = 0.65 = 65%
```

2. Annualized Return

```
Annualized Return = (1 + Cumulative Return)^(1/Years) - 1

Example:
65% over 5 years:
Annualized = (1.65)^(1/5) - 1 = 1.1052 - 1 = 10.52% per year
```

3. Annualized Volatility

```
Daily Volatility = StdDev(Daily Returns)
Annualized Volatility = Daily Volatility × √252

Example:
Daily StdDev = 1.2%
Annualized Vol = 1.2% × √252 = 1.2% × 15.87 = 19.05%
```

4. Win Rate

```
Win Rate = Number of Positive Days / Total Days × 100

Example:
Positive days = 650, Total days = 1200
Win Rate = 650/1200 = 54.2%
```

5. Transaction Costs

```
Transaction Cost = Trade Value × Cost Rate
Cost Rate = 10 basis points = 0.10% = 0.001

Example:
Buying $10,000 worth of AAPL:
Cost = $10,000 × 0.001 = $10
```

Step-by-Step Code Explanation

The Main File: `main.py`

```python
# STEP 1: Get the data (download from Yahoo Finance)
collector = DataCollector(years=5)
collector.download_all()
prices = collector.get_combined_prices()  # DataFrame: rows=dates, cols=tickers

# STEP 2: Generate signals
signals_gen = AlphaSignals()
signals = signals_gen.generate_all_signals(prices)
# signals = {'momentum': DataFrame, 'mean_reversion': DataFrame, 'volatility': DataFrame}

combined = signals_gen.combine_signals(signals)
# combined = DataFrame with final scores for each stock each day

# STEP 3: Build portfolio
pc = PortfolioConstructor(n_long=3, n_short=3)
long_tickers, short_tickers = pc.select_positions(combined.iloc[-1])
```

```python
# long_tickers = ['NVDA', 'AAPL', 'MSFT']
# short_tickers = ['TSLA', 'META', 'WMT']

# STEP 4: Run backtest (simulate trading over 5 years)
backtester = Backtester(
    initial_capital=100000,      # Start with $100K
    transaction_cost_bps=10,     # 0.10% per trade
    risk_free_rate=0.02          # 2% Treasury rate
)
result = backtester.run(prices, benchmark_returns)

# STEP 5: Analyze performance
analyzer = PerformanceAnalyzer()
report = analyzer.generate_summary_report(result)

# STEP 6: (Optional) Optimize signal weights
optimizer = StrategyOptimizer(
    training_months=12,   # Train on 1 year
    validation_months=3,  # Test on 3 months
    step_months=3         # Move forward 3 months and repeat
)
opt_result = optimizer.walk_forward_optimize(prices)
# Finds optimal weights: Momentum=61%, MeanRev=23%, Volatility=16%
```

Each Module Explained

1. `data_collector.py` - The Data Fetcher

Purpose: Download stock prices from Yahoo Finance

Key Code:

```python
DEFAULT_TICKERS = ['AAPL', 'MSFT', 'AMZN', 'GOOGL', 'META',
                   'TSLA', 'NVDA', 'JPM', 'JNJ', 'WMT']
BENCHMARK = '^GSPC'  # S&P 500

# Download data
stock = yf.Ticker('AAPL')
df = stock.history(start='2021-01-01', end='2026-01-01')
# Returns: Date, Open, High, Low, Close, Volume
```

---

2. `alpha_signals.py` - The Stock Ranker

Purpose: Calculate the 3 signals and combine them

Key Code:

```
# Momentum: 20-day returns
returns = prices.pct_change(20)

# Mean Reversion: deviation from 50-day MA
ma = prices.rolling(50).mean()
deviation = (prices - ma) / ma
mean_reversion = -deviation  # Invert

# Volatility: 30-day rolling, annualized
volatility = returns.rolling(30).std() * np.sqrt(252)
inv_volatility = -volatility  # Invert (lower = better)

# Combine with weights
combined = (momentum * 0.61) + (mean_reversion * 0.23) + (volatility * 0.16)
```

---

3. `portfolio.py` - The Portfolio Builder

Purpose: Pick stocks and calculate weights

Key Code:

```
# Sort by combined score
sorted_scores = alpha_scores.sort_values(ascending=False)

# Top 3 = LONG
long_tickers = sorted_scores.head(3).index.tolist()

# Bottom 3 = SHORT
short_tickers = sorted_scores.tail(3).index.tolist()

# Calculate weights
if market_regime == 'BULL':
    long_weight = 1.0 / 3  # 33.3% each
    short_weight = 0
else:
    long_weight = 0.5 / 3   # 16.67% each
    short_weight = -0.5 / 3  # -16.67% each
```

---

4. `backtester.py` - The Time Machine

Purpose: Simulate trading over 5 years

Key Code:

```python
for date in trading_dates:
    # Update prices
    current_prices = prices.loc[date]

    # Check stop-loss (-10%)
    for ticker, position in positions.items():
        if position.return_pct <= -0.10:
            close_position(ticker)  # SELL immediately!

    # Rebalance monthly
    if date in rebalance_dates:
        # Sell old positions
        # Buy new positions based on new signals
        rebalance(target_weights, current_prices)

    # Track portfolio value
    portfolio_value = cash + sum(position_values)
```

---

5. `risk_manager.py` - The Safety Guard

Purpose: Protect from big losses

Key Code:

```python
# Stop-loss check
def check_stop_loss(position):
    if position.is_long:
        return_pct = (current_price / entry_price) - 1
    else:
        return_pct = (entry_price / current_price) - 1

    return return_pct <= -0.10  # True if loss > 10%

# Dollar-neutral check
long_exposure = sum(long_position_values)   # e.g., $50,000
short_exposure = sum(short_position_values)  # e.g., $50,000
is_balanced = abs(long_exposure - short_exposure) < tolerance
```

---

6. `performance.py` - The Report Card

Purpose: Calculate metrics and create charts

Key Calculations:

```python
# Cumulative return
cumulative_return = (portfolio_values.iloc[-1] / initial_capital) - 1

# Sharpe ratio
excess_return = annualized_return - 0.02
sharpe_ratio = excess_return / annualized_volatility

# Max drawdown
rolling_max = portfolio_values.expanding().max()
drawdowns = (portfolio_values - rolling_max) / rolling_max
max_drawdown = drawdowns.min()

# Alpha & Beta
covariance = np.cov(strategy_returns, benchmark_returns)[0, 1]
variance = np.var(benchmark_returns)
beta = covariance / variance
alpha = strategy_return - (risk_free + beta * (market_return - risk_free))
```

---

7. `optimizer.py` - The Weight Finder

Purpose: Find optimal signal weights using Walk-Forward Analysis

Process:

```
|-------- Training (12 months) --------|--- Test (3 months) ---|

Step 1: Train on months 1-12, test on months 13-15
Step 2: Train on months 4-15, test on months 16-18
Step 3: Train on months 7-18, test on months 19-21
... continue until end of data ...

Final: Average all optimal weights found
Result: Momentum=61%, Mean Reversion=23%, Volatility=16%
```

Key Code:

```python
# Grid search over weight combinations
for momentum_weight in [0, 0.2, 0.4, 0.6, 0.8, 1.0]:
    for meanrev_weight in [0, 0.2, 0.4, 0.6, 0.8, 1.0]:
        volatility_weight = 1.0 - momentum_weight - meanrev_weight

        if volatility_weight >= 0:
            sharpe = calculate_sharpe(weights)
            if sharpe > best_sharpe:
                best_weights = weights
```

Complete Worked Example

Day 1: January 1, 2021

Starting Capital: \$100,000

Step 1: Download prices

```
AAPL: $130    MSFT: $220    AMZN: $3200   GOOGL: $1750
META: $270    TSLA: $700    NVDA: $130    JPM: $125
JNJ: $155     WMT: $145
```

Step 2: Calculate signals (after 50 days of data)

Stock	Momentum	Mean Rev	Volatility	Combined
NVDA	+1.20	+0.30	-0.20	**+0.81**
AAPL	+0.80	+0.10	+0.30	**+0.58**
MSFT	+0.60	+0.20	+0.20	**+0.44**
GOOGL	+0.20	+0.10	+0.10	+0.16
JPM	+0.10	+0.00	+0.40	+0.13
JNJ	-0.10	+0.20	+0.50	+0.06
WMT	-0.30	+0.10	+0.20	-0.13
AMZN	-0.40	-0.10	-0.10	-0.29
META	-0.60	-0.30	-0.30	**-0.50**
TSLA	-0.80	-0.20	-0.80	**-0.65**

Step 3: Select positions

```
LONG: NVDA, AAPL, MSFT (top 3)
SHORT: AMZN, META, TSLA (bottom 3)
```

Step 4: Calculate weights (assuming neutral regime)

```
Long: $50,000 / 3 = $16,667 each
Short: $50,000 / 3 = $16,667 each

Positions:
  NVDA: Buy 128 shares × $130 = $16,640 LONG
  AAPL: Buy 128 shares × $130 = $16,640 LONG
  MSFT: Buy 75 shares × $220 = $16,500 LONG

  AMZN: Short 5 shares × $3200 = $16,000 SHORT
  META: Short 61 shares × $270 = $16,470 SHORT
  TSLA: Short 23 shares × $700 = $16,100 SHORT
```

Step 5: After 1 month, prices changed

```
NVDA: $130 → $145 (+11.5%)
AAPL: $130 → $135 (+3.8%)
```

```
MSFT: $220 → $230 (+4.5%)
TSLA: $700 → $650 (-7.1%)  ← Our short made money!
META: $270 → $260 (-3.7%)  ← Our short made money!
AMZN: $3200 → $3100 (-3.1%)  ← Our short made money!
```

P&L Calculation:

```
LONG P&L:
  NVDA: 128 × ($145 - $130) = +$1,920
  AAPL: 128 × ($135 - $130) = +$640
  MSFT: 75 × ($230 - $220) = +$750
  Total Long P&L = +$3,310

SHORT P&L:
  TSLA: 23 × ($700 - $650) = +$1,150
  META: 61 × ($270 - $260) = +$610
  AMZN: 5 × ($3200 - $3100) = +$500
  Total Short P&L = +$2,260

TOTAL P&L = +$3,310 + $2,260 = +$5,570

Transaction Costs (buying & rebalancing):
  ~$200 total

NET P&L = +$5,370
New Portfolio Value = $105,370
Monthly Return = 5.37%
```

How to Explain to Others

30-Second Pitch:

> "I built an automated stock trading strategy that picks winners and losers from 10 major US stocks. It buys the top 3 and shorts the bottom 3 every month. It made 65% over 5 years while managing risk carefully with stop-losses and staying market-neutral."

2-Minute Explanation:

> "This is a 'long-short' quantitative trading strategy. Here's how it works:
>
> First, I get 5 years of price data for 10 big companies.
>
> Then I calculate 3 'signals' for each stock: - **Momentum**: Is it going up? Trending stocks keep trending. - **Mean Reversion**: Did

it drop too much? It might bounce back. - **Volatility**: Is it stable? I prefer calm stocks.

I combine these using weights I optimized: 61% momentum, 23% mean reversion, 16% volatility.

Based on combined scores, I rank all 10 stocks. I BUY the top 3, SHORT the bottom 3.

The strategy is 'dollar-neutral' - equal money on both sides. If the market crashes, my shorts profit while longs lose. I'm protected!

Every month, I recalculate and rebalance. If any position drops 10%, my stop-loss triggers.

The formulas are standard finance: Sharpe ratio for risk-adjusted returns, z-scores for normalization, moving averages for trends.

In backtesting, it beat the S&P 500 with lower risk!"

Key Numbers to Remember:

- **10 stocks** in universe
- **3 long, 3 short** positions
- **61% momentum, 23% mean reversion, 16% volatility** weights
- **Monthly** rebalancing
- **10%** stop-loss
- **10 basis points** (0.10%) transaction cost
- **5 years** of historical data

Common Questions & Answers

Q: What is alpha? A: Alpha is the extra return above what you'd expect given the market risk. If the market returned 10% and I returned 15%, my alpha is roughly 5%.

Q: Why 61% weight for momentum? A: Through walk-forward optimization (testing on historical data), I found momentum was the most predictive signal for this stock universe.

Q: What's the Sharpe ratio formula? A: Sharpe = (Return - Risk-Free Rate) / Volatility. It measures return per unit of risk.

Q: How does shorting work? A: You borrow shares from a broker, sell them immediately, wait for price to drop, buy back cheaper, return shares, keep the profit.

Q: What's dollar-neutral? A: Equal money long and short. If market crashes 20%, your longs lose ~20% but your shorts gain ~20%. Net = roughly zero loss.

Q: Did this use real money? A: No, this is backtesting - simulating on historical data. No real trades were made.

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Project: https://github.com/digantk31/Alpha-Long-Short
Date: January 2026